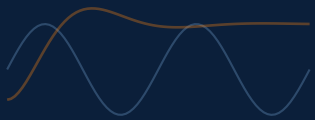
WEEK 1 · DAY 1 / 5

Mechatronics and Robotics Engineering (50 h)



Electrical Machines and Actuators in Robots I

Energy conversion and the actuator landscape · DC, BLDC and PMSM motors · motor models and time constants · stepper and induction motors

Mechatronics and Robotics Engineering Programme

Week 1 · Day 1 ▪ 2 weeks · 10 days · 50 hours

Agenda and learning outcomes

Code	Session	Time
1.1	Electromechanical energy conversion & the actuator landscape in robots	1.0 h
1.2	DC, BLDC and PMSM motors: structure and commutation	1.5 h
1.3	Electrical and mechanical motor models	1.5 h
1.4	Stepper and induction motors in automation	1.0 h
	Net teaching time	**5.0 h**

Breaks (≈10 min) roughly every hour are additional.

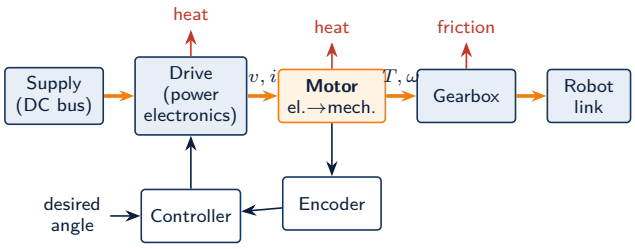
By the end of today you can . . .

- **explain** how a motor turns current into torque
- **compare** electric, pneumatic and hydraulic actuators for a robot
- **sketch** how DC, BLDC and PMSM motors commutate
- **write** the DC-motor equations and **calculate** its time constants
- **choose** between stepper, induction and servo motors

Session 1

1.1 Energy conversion & the actuator landscape

A robot joint is a chain of energy converters



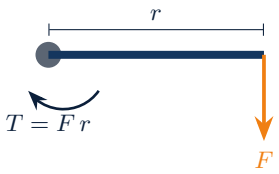
- Each joint of a 6-axis arm repeats this chain: **6 motors, 6 drives, 6 encoders**.
- The **actuator** is the part that turns energy into motion.
- Energy flows right; information (angle) flows back to the controller.
- Every block loses some energy as **heat**.



Oleg Yunakov, CC BY-SA 4.0, via Wikimedia Commons

Industrial arm: one motor + gearbox per joint.

Refresher: torque, speed and power



spanner on a bolt = robot link on a joint

Linear	Rotation	Unit
force F	torque T	N m
velocity v	angular speed ω	rad/s
$P = F\,v$	$P = T\,\omega$	W

Analogy: bicycle gears – the same pedalling power can be "lots of force, slowly" or "little force, fast".

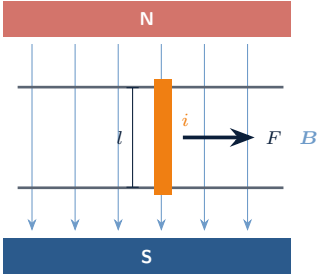
$$P = T\,\omega, \qquad \omega = \frac{2\pi n}{60} \quad (n \text{ in rpm})$$

Worked example: one robot joint

1. Joint needs $T = 50$ N m at $\omega = 1.05$ rad/s ($60°$/s).
2. Power: $P = 50 \cdot 1.05 \approx 52$ W.
3. A motor at 1000 rpm (105 rad/s) gives this with only $T = 52/105 \approx 0.5$ N m – through a 100:1 gearbox.

Take-away: power is the product; gears only trade torque for speed.

Refresher: magnetic field and the force on a wire



wire on two rails between magnet poles

- A magnet fills space with a **magnetic field $\boldsymbol{B}$** (unit: tesla, T). Picture it as arrows from N to S.
- A wire carrying current through that field is **pushed sideways** – perpendicular to both i and $\boldsymbol{B}$.
- Direction: right-hand rule (fingers along i, bend toward $\boldsymbol{B}$, thumb = force).

$$F = B\,l\,i \qquad F \text{ [N]},\ B \text{ [T]},\ l \text{ [m]},\ i \text{ [A]}$$

Example: $B = 0.5\,\text{T}$, $l = 0.1\,\text{m}$, $i = 4\,\text{A} \Rightarrow F = 0.2\,\text{N}$.

Take-away: current + magnetic field = force. Every electric motor uses this.

Motor action and generator action: two halves of every motor

1. Motor action (Lorentz): current in a field gives force, $F = Bli$.

2. Generator action (Faraday): a wire **moving** at speed v through the field gets a voltage $e = Blv$, the **back-EMF**.

Both happen at the same time in every motor.

Bend the wire into a coil (N turns, radius r) and both laws use the same constant $k = 2NBlr$:

$$T = k_t\, i, \qquad e = k_e\, \omega, \qquad k_t = k_e \text{ (SI units)}$$

Worked example

Coil: $N = 50$, $B = 0.5\,\text{T}$, $l = 0.04\,\text{m}$, $r = 0.02\,\text{m}$

1. $k = 2 \cdot 50 \cdot 0.5 \cdot 0.04 \cdot 0.02 = 0.04\,\text{N m/A}$
2. At $i = 5\,\text{A}$: $T = 0.04 \cdot 5 = 0.2\,\text{N m}$
3. At $\omega = 300\,\text{rad/s}$: $e = 0.04 \cdot 300 = 12\,\text{V}$

Take-away

Torque constant k_t (N m/A) and **back-EMF constant** k_e (V s/rad) are the same number: **current makes torque, speed makes voltage**.

Reluctance force: iron is pulled into the field



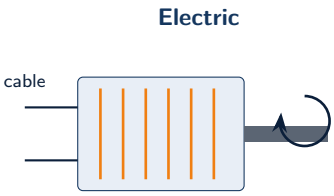
Analogy: a fridge magnet snaps onto the steel door – the iron moves to where the magnetic path is **easiest**.

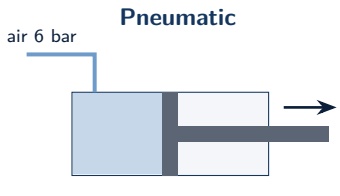
- No magnet needed on the rotor: a plain iron bar turns to line up with the coil.
- Torque pushes towards the position of **highest inductance** L.
- Torque grows with i^2: current direction does not matter.
- Used in solenoids, **stepper motors** (Session 1.4) and switched-reluctance motors.

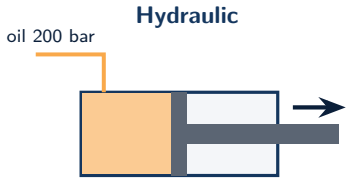
$$T = \tfrac{1}{2}\, i^2\, \frac{dL}{d\theta} \qquad (L \text{ in H}, \theta \text{ in rad})$$

Going further: the co-energy method $T = \partial W'/\partial\theta$ also covers saturated iron [Fitzgerald et al., 2003, ch. 3].

The actuator landscape: three ways to move a robot







- **Electric**: energy arrives by cable; easiest to **control precisely**.
- **Pneumatic**: compressed air pushes a piston; cheap, fast, **springy**.
- **Hydraulic**: oil at high pressure; **huge force** from a small cylinder.

> Today's industrial and collaborative robots are almost all **electric**. Pneumatics live in grippers and simple pick-and-place; hydraulics in heavy and legged machines.

Fluid power in numbers: force from pressure



piston area $A = \pi d^2/4$



Jonashtand, CC BY-SA 4.0, via Wikimedia Commons

$F = p\,A$ (N, Pa, m^2); 1 bar $= 10^5$ Pa

Worked example: bore 40 mm

1. $A = \pi \cdot 0.04^2/4 = 1.26 \times 10^{-3}$ m^2
2. Pneumatic, 6 bar: $F = 6 \times 10^5 \cdot 1.26 \times 10^{-3} \approx 750$ N
3. Hydraulic, 200 bar: $F \approx 25$ kN – **33× more**

Air is springy: **hard to stop mid-stroke**. Oil is stiff, but needs pumps and hoses.

Comparing actuator technologies for robots

	Electric	Pneumatic	Hydraulic
Force per size	medium–high	low	**very high**
Positioning accuracy	**very good**	poor (2 end stops)	good
Control	**easy, digital**	on/off valves	servo valves
Efficiency	**80–95 %**	10–20 %	40–60 %
Cost / infrastructure	medium	**low**	pump, hoses, leaks
Typical robot use	arm joints, AMRs	grippers, suction	heavy, legged robots



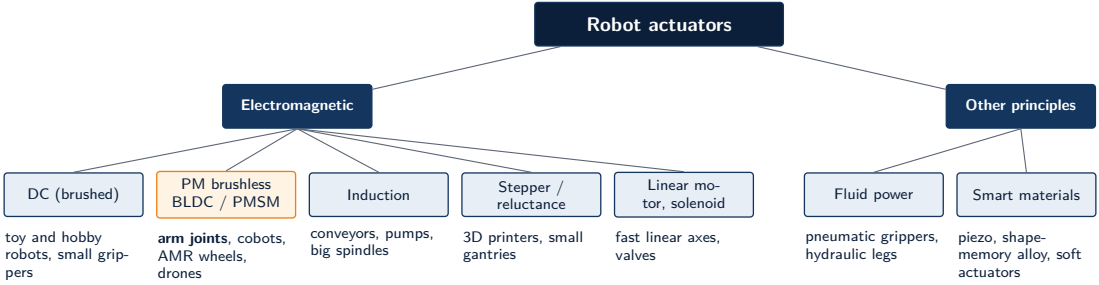
S.J. de Waard, CC BY 2.5, via Wikimedia Commons

Hydraulic cylinders: large forces in heavy machines.

Why robots went electric

Clean, efficient and easy to control with a computer and an encoder; gearboxes (Day 2) fix their weakness of low torque.

Family tree of electromagnetic actuators



- Today: the **rotary electromagnetic** branch (Sessions 1.2–1.4).
- Day 2: gearboxes, linear actuators, grippers, sizing.

Going further: soft robots use pneumatic "muscles" and electro-active polymers – high compliance, low precision.

Session 1.1 – Check your understanding

Check your understanding

1. A motor has $k_t = 0.1\,\mathrm{N\,m/A}$. You double the current from 2 A to 4 A. What happens to the torque?
2. A joint delivers 2 N m at 1500 rpm. Roughly how much mechanical power is that?
3. Why is a pneumatic cylinder a poor choice when a robot must stop accurately halfway along its stroke?

How to work:

- 2 minutes alone, then compare with your neighbour.
- Write the formula first, then the numbers with units.
- Hint: $T = k_t i$, $P = T\omega$, $F = pA$.

Session 1.1 – Summary and key terms

- A motor turns $v\,i$ into $T\,\omega$ through a magnetic field – and back again when braking.
- Current in a field gives force; motion in a field gives voltage.
- Iron alone is also pulled into a field (reluctance).
- Electric actuators dominate robots; pneumatics for grippers, hydraulics for heavy loads.

Remember

$$P = T\omega, \quad T = k_t\, i, \quad e = k_e\, \omega, \quad F = p\, A$$

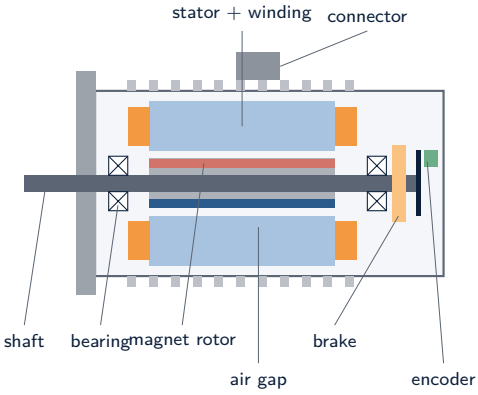
Key terms

Actuator: device that turns energy into motion.
Torque T: turning effect of a force, N m.
Back-EMF: voltage a spinning motor generates.
Torque constant k_t: torque per ampere.
Reluctance: "difficulty" of the magnetic path.
Efficiency η: power out / power in.

Session 2

1.2 DC, BLDC and PMSM motors

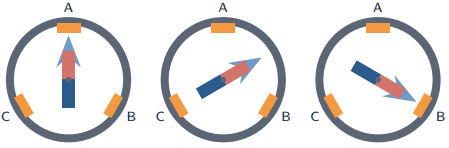
Anatomy of a robot servo motor



Schematic cut through a PM servo motor (not to scale).

- **Stator**: fixed outer part – iron stack with copper windings. Gets hot, so it touches the housing.
- **Rotor**: turning part – shaft with magnets. Long and slim, so it is light to accelerate.
- **Air gap**: thin gap ($< 1\,\text{mm}$) where torque is made.
- **Encoder**: measures the rotor angle (Week 2).
- **Brake**: holds the arm when power is off – vital on a robot, or the arm falls.

Refresher: a magnet follows a rotating field



Stator field (blue arrow) moves step by step; the magnet rotor (red = N) follows.

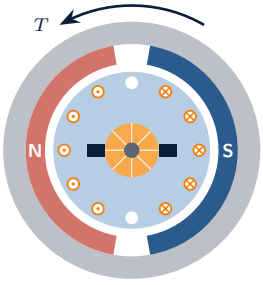
Analogy: a compass needle next to a coil – switch coils around the circle and the needle chases the field like a donkey chasing a carrot.

- Three coils A, B, C, 120° apart, make a field we can **point anywhere** by choosing their currents.
- The rotor magnet always tries to line up with it.
- Torque is largest when field and magnet are 90° **apart**, zero when aligned.

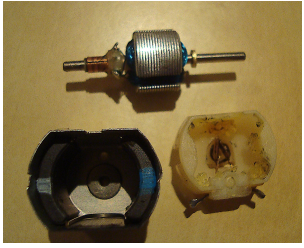
$T \propto B_{\text{stator}}\, B_{\text{rotor}} \sin\delta$ (δ: angle between them)

Take-away: commutation = keep the field 90° ahead of the rotor.

Brushed DC motor: construction



magnets, armature, commutator

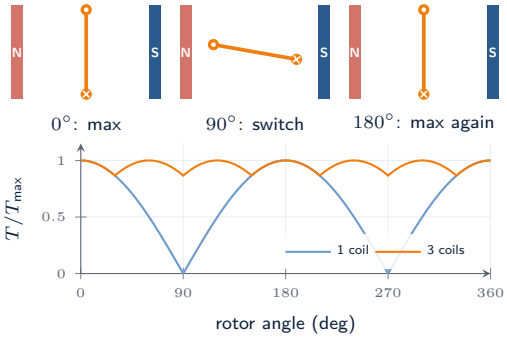


Jfer91, CC BY-SA 3.0, via Wikimedia Commons

Opened DC motor: magnets in the can, armature and commutator on the shaft.

- **Stator magnets** make the field.
- **Armature**: the coils on the **rotor** that carry the current.
- **Commutator + brushes**: a rotating switch that feeds the coils.
- Simple: connect a battery and it spins; speed $\approx$ proportional to voltage.
- Found in toy robots, small grippers, car wipers.

How the commutator keeps the torque in one direction

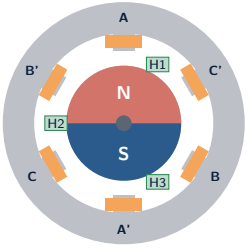


- Without switching, the coil would rock back and forth: torque changes sign every half turn.
- The **commutator** reverses the coil current exactly when the coil passes the "dead" position.
- Result: torque always pushes **the same way** (rectified $|\cos|$).
- More coils and segments $\Rightarrow$ smaller dips $=$ less **torque ripple**.

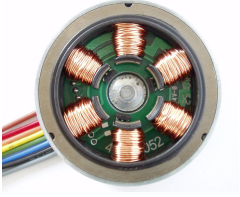
Weak points of brushes

Brush wear and sparks, heat trapped in the rotor, speed limit. **Idea:** swap roles – magnets on the rotor, coils on the stator, switches made of transistors.

Brushless DC (BLDC): the “inside-out” DC motor



3 phases A, B, C + Hall sensors H1–H3

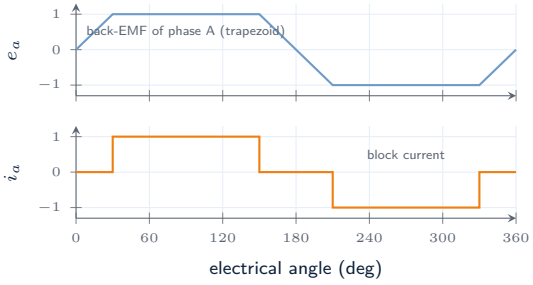


Medvedev, CC BY-SA 3.0, via Wikimedia Commons

- **Magnets on the rotor**, three-phase coils on the stator (photo) ⇒ no brushes, no wear.
- Heat is made in the stator, right next to the housing ⇒ **easy cooling**.
- The drive must switch the coils itself, so it needs to know the rotor angle: three cheap **Hall sensors**.
- This is **electronic commutation**: transistors replace the commutator.

Same physics as the brushed motor ($T = k_t i$, $e = k_e \omega$) – only the switch moved from copper segments into the drive.

BLDC six-step commutation

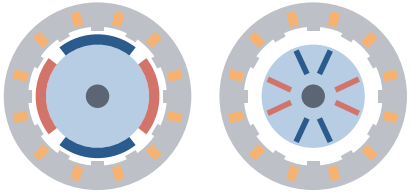


Hall H1 H2 H3	+ phase	− phase
101	A	B
100	A	C
110	B	C
010	B	A
011	C	A
001	C	B

Example sequence; the exact table depends on sensor placement.

- At any moment **two phases conduct**, one rests.
- 6 steps per electrical turn, each 60°.
- Simple and cheap, but a small **torque ripple** at every step.

PMSM: smooth sine currents – the robot servo motor



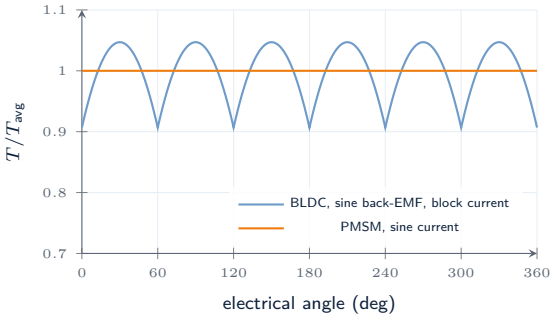
- **PMSM** (permanent-magnet synchronous motor): same build as BLDC, but sine-shaped back-EMF.
- An **encoder** gives the exact angle; the drive feeds three smooth **sine currents** 120° apart.
- The field turns smoothly $\Rightarrow$ almost **ripple-free torque**.
- Industrial "AC servo motors" and almost every robot arm joint use a PMSM.



Sebastian Koppehel, CC BY 3.0, via Wikimedia Commons

Going further: field-oriented control computes these sine currents from the rotor angle: $T = \frac{3}{2}\, p\, \psi_{\mathrm{PM}}\, i_q$ (W1D5).

Torque ripple: six-step versus sinusoidal



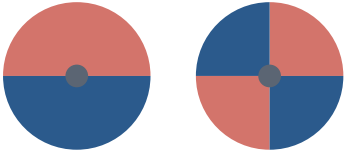
Ideal case. With block currents on a sine-EMF motor the dips reach $\approx 13\,\%$ every 60°.

Analogy: a bicycle pedalled with jerky pushes vs. smooth circles – same average power, different feel.

- **Torque ripple**: periodic wobble on top of the mean torque.
- In a robot it shows up as **vibration** at the tool tip and poor slow-speed motion.
- Sine commutation with an accurate encoder removes most of it.

Drones and fans: BLDC six-step is fine. Robot arms painting or welding: PMSM.

Pole pairs: electrical versus mechanical angle



$p = 1$ (2 poles) $p = 2$ (4 poles)

One N–S pair passing a coil = one **electrical** cycle (360°_{el}).

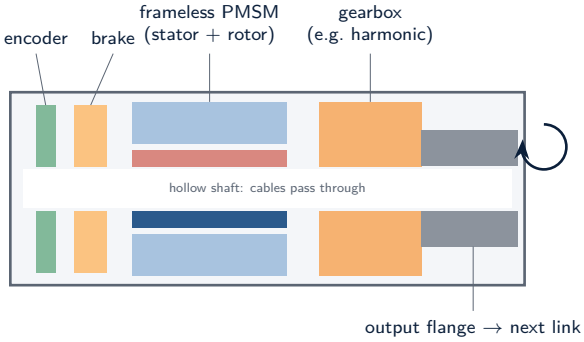
$$\theta_{\text{el}} = p\,\theta_{\text{mech}}, \qquad \omega_{\text{el}} = p\,\omega_{\text{mech}}, \qquad f_{\text{el}} = \frac{p\,n}{60}$$

Worked example: robot joint motor

1. PMSM with $p = 4$ pole pairs (8 poles) at $n = 3000$ rpm.
2. Mechanical: $3000/60 = 50$ rev/s.
3. Electrical: $f_{\text{el}} = 4 \cdot 50 = 200$ Hz.
4. The drive must make 200 Hz sine currents.

- More poles ⇒ more torque from a flat, short motor (good for robot joints).
- But higher electrical frequency ⇒ faster drive needed.

Inside a robot joint: frameless motor, brake, encoder, gear



- **Frameless motor**: only stator ring and rotor ring are bought; the robot housing is the motor housing.
- **Hollow shaft**: cables run through the joint.
- Order: encoder → brake → motor → gearbox → link.
- Compact, light, integrated: typical of **cobots**.

Going further: many cobot joints add a second encoder on the output side to measure the gearbox twist (W2D4 sensors).

Comparison: DC, BLDC and PMSM in robots

	Brushed DC	BLDC	PMSM (AC servo)
Commutation	brushes + commutator	6 steps, Hall sensors	sine currents, encoder
Coils on	rotor	stator	stator
Torque per size	medium	high	**very high**
Smoothness	medium	ripple at each step	**very smooth**
Maintenance	brush wear	none	none
Drive electronics	**simplest**	simple	most complex
Robot examples	toy robots, small grippers	drones, AMR wheels, fans	arm joints, cobots, CNC axes

Rule of thumb

Need precise, smooth, dynamic motion? $\Rightarrow$ **PMSM**. Need cheap and simple? $\Rightarrow$ brushed DC or BLDC. BLDC and PMSM are built almost the same – the difference is mainly **how the drive feeds them**.

Session 1.2 – Check your understanding

Check your understanding

1. What job does the commutator do in a brushed DC motor – and what does that job in a BLDC motor?
2. Why is a brushless motor easier to cool than a brushed motor of the same size?
3. A PMSM with $p = 5$ pole pairs runs at 1200 rpm. What electrical frequency must the drive produce?

Hint: think about *where* the current flows (rotor or stator), *who* knows the rotor angle, and $f_{\text{el}} = p\,n/60$.

Session 1.2 – Summary and key terms

- Every motor: stator, rotor, air gap; torque is largest with field and magnet 90° apart.
- Coil currents must be switched as the rotor turns (**commutation**).
- Brushed DC: mechanical switch. BLDC: 6 steps + Hall. PMSM: sine + encoder.
- More pole pairs: more torque, higher electrical frequency.

Remember

$\omega_{\text{el}} = p\,\omega_{\text{mech}}, \qquad f_{\text{el}} = p\,n/60$

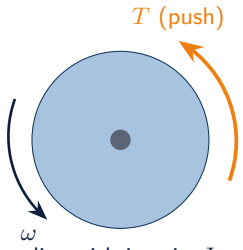
Key terms

Stator / rotor: fixed / turning part.
Commutation: switching coil currents with rotor angle.
Hall sensor: detects magnet polarity (on/off).
Torque ripple: periodic torque wobble.
Pole pair p: one N + one S magnet.
Frameless motor: stator + rotor without housing.

Session 3

1.3 Electrical and mechanical motor models

Refresher: Newton's law for rotation



disc with inertia J

Linear	Rotation	Unit
mass m	inertia J	$\mathrm{kg\,m^2}$
$F = m\,\dfrac{dv}{dt}$	$T = J\,\dfrac{d\omega}{dt}$	

Analogy: a heavy playground roundabout is hard to start and hard to stop – it has a large **moment of inertia** J.

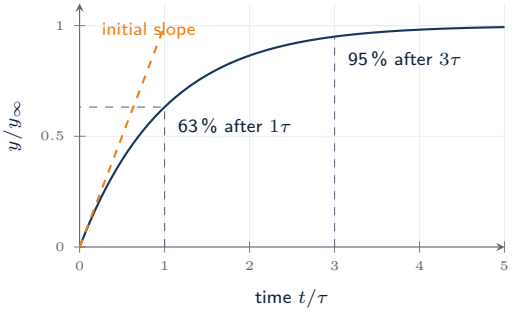
$$J\,\frac{d\omega}{dt} = \sum T = T_{\text{motor}} - T_{\text{friction}} - T_{\text{load}}$$

Worked example

1. Rotor $J = 1 \times 10^{-4}\,\mathrm{kg\,m^2}$, net torque $T = 0.5\,\mathrm{N\,m}$.
2. Acceleration $\alpha = T/J = 5000\,\mathrm{rad/s^2}$.
3. Time to reach $314\,\mathrm{rad/s}$ (3000 rpm): $t = 314/5000 = 0.063\,\mathrm{s}$.

Disc: $J = \frac{1}{2}mr^2$. **Take-away:** net torque sets how fast the speed changes.

Refresher: first-order ODEs and the exponential response



You know this from RC charging: the capacitor voltage rises fast at first, then slower and slower.

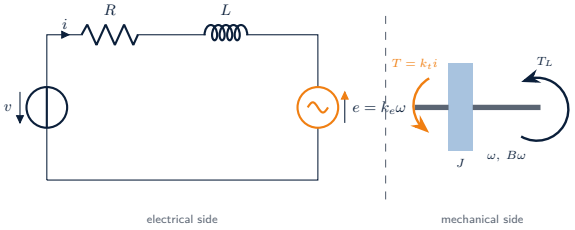
$$\tau \frac{dy}{dt} + y = y_\infty \;\Rightarrow\; y(t) = y_\infty\left(1 - e^{-t/\tau}\right)$$

- τ = **time constant** [s]: RC circuit $\tau = RC$, RL circuit $\tau = L/R$.
- After 1τ: 63 %; after 3τ: 95 %; after 5τ: "done".

Example: $L = 0.5\,\text{mH}$, $R = 1\,\Omega \Rightarrow \tau = 0.5\,\text{ms}$; current settles in $\approx 1.5\,\text{ms}$.

Take-away: one energy store $\Rightarrow$ one τ $\Rightarrow$ one exponential.

Equivalent circuit of the DC motor



The electrical side is the **RL circuit** you know, plus one extra "battery" $e = k_e\omega$ that grows with speed and **opposes** the supply.

Sym.	Meaning	Unit
R	winding resistance	Ω
L	winding inductance	H
k_e	back-EMF constant	V s/rad
k_t	torque constant	N m/A
J	rotor (+ load) inertia	kg m^2
B	viscous friction	N m s/rad
T_L	load torque	N m

Signs: v, i positive into the motor; T_L works against the motion.

BLDC/PMSM: same model per phase (W1D5).

The two motor equations

Electrical (KVL around the loop):

$$v = R\,i + L\frac{di}{dt} + k_e\,\omega$$

Mechanical (Newton for rotation):

$$J\frac{d\omega}{dt} = k_t\,i - B\,\omega - T_L$$

- Linked both ways: i makes torque, ω makes back-EMF.
- Two first-order ODEs $\Rightarrow$ one **second-order** system.

Why the two constants are equal

Power taken from the circuit by the back-EMF: $e\,i = k_e\,\omega\,i$
Power delivered to the shaft: $T\,\omega = k_t\,i\,\omega$
Energy is conserved $\Rightarrow$ $\boldsymbol{k_t = k_e}$ (SI units).

Datasheet units: k_e is often given in V per 1000 rpm. Divide by 104.7 to get V s/rad: $10.47\,\text{V/krpm} \Rightarrow k_e = k_t = 0.1$ SI.

Going further: three-phase datasheets use rms and line-to-line values, so k_t and k_e differ by factors like $\sqrt{3}$; resistance rises $\approx$0.4 %/K when hot.

Reading a motor datasheet: worked example

Datasheet value (example)	
Rated voltage v_N	48 V
Rated torque T_N	0.64 N m
Rated speed n_N	3000 rpm
Torque constant k_t	0.12 N m/A
Terminal resistance R	0.8 Ω
Terminal inductance L	1.2 mH
Rotor inertia J_m	0.6×10^{-4} kg m^2

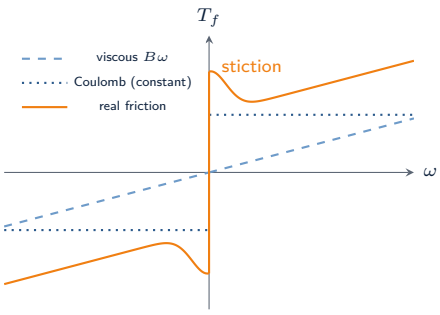
Illustrative values for a 200 W class motor.

What can we compute?

1. Rated current: $i_N = T_N/k_t = 0.64/0.12 \approx 5.3$ A.
2. Back-EMF at 3000 rpm (314 rad/s): $e = 0.12 \cdot 314 \approx 38$ V.
3. Voltage needed: $v = Ri + e = 0.8 \cdot 5.3 + 38 \approx 42\,\text{V} < 48\,\text{V}$ ✓
4. Copper loss: $P_{\text{Cu}} = Ri^2 = 0.8 \cdot 5.3^2 \approx 23$ W of heat.

Output power: $T\omega = 0.64 \cdot 314 \approx 200$ W.
Take-away: back-EMF uses most of the supply voltage at full speed.

Friction and load torque



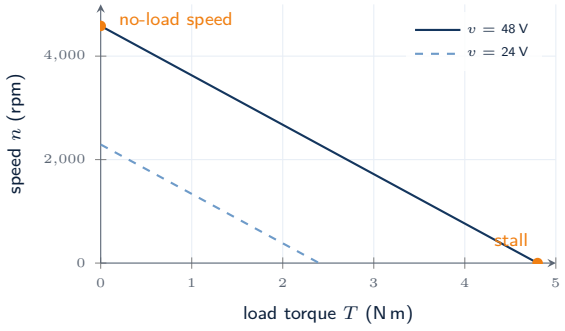
Analogy: pushing a heavy box – hard to get it moving, easier once it slides.

- **Viscous** friction $B\omega$: grows with speed, like stirring honey.
- **Coulomb** friction: constant, flips sign with direction.
- **Stiction**: extra "sticking" torque at standstill $\Rightarrow$ jerky start.
- **Load torque** T_L: gravity on a robot arm, payload, contact forces.

model used today: $T_f \approx B\,\omega$ (linear, easy maths)

[Armstrong-Hélouvry, Dupont & Canudas de Wit, 1994]

Steady state: the speed–torque line



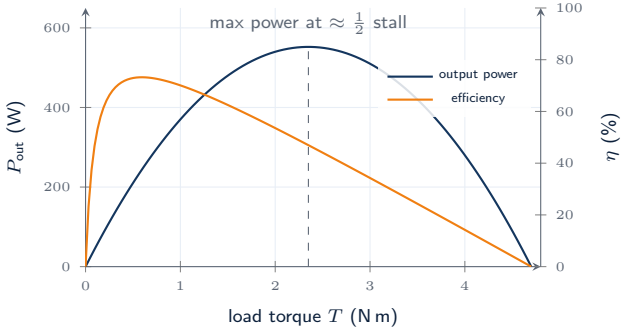
Example motor: $R = 1\,\Omega$, $k_t = k_e = 0.1$ SI, friction neglected.

Constant speed: set $\frac{d}{dt} = 0$ in both equations:

$$\omega = \frac{v}{k_e} - \frac{R}{k_e k_t} T$$

1. No load: $\omega_0 = 48/0.1 = 480$ rad/s (4584 rpm).
2. Stalled: $i = v/R = 48$ A, $T = k_t i = 4.8$ N m.
3. More load $\Rightarrow$ speed **droops** along the line.
4. Higher voltage $\Rightarrow$ line moves up.

Output power and efficiency along the line



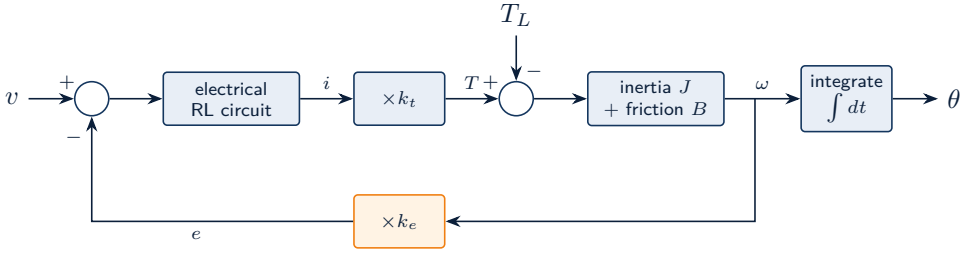
Same motor at 48 V, with 0.1 N m friction torque.

- $P_{\text{out}} = T\,\omega$ is a **parabola**: zero at no load (no torque) and at stall (no speed).
- Efficiency is **highest at light load**; it falls as copper loss Ri^2 grows.
- Real motors are rated far below stall – heat (not the line) sets the limit.

$$\eta = \frac{T\,\omega}{v\,i}, \qquad \text{losses} = v\,i - T\omega \rightarrow \text{heat}$$

Going further: rated (continuous) torque is set by winding temperature; peak torque by magnet demagnetisation and drive current (W1D2 sizing).

Block diagram of the DC motor

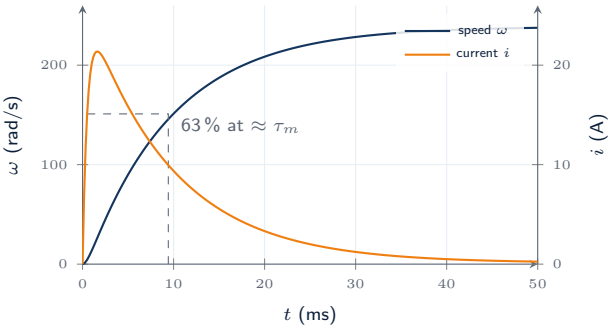


- A **block diagram** draws the equations as signal flow: boxes act on signals, circles add or subtract.
- Back-EMF is a built-in **feedback loop**: faster ⇒ more e ⇒ less current ⇒ less torque.
- T_L (gravity, payload) enters as a **disturbance**.

Robot drives control the **current** directly, so $T \approx k_t\, i^*$: the motor behaves like a **torque source** (W1D4–W1D5).

Going further: with the Laplace transform each box becomes a transfer function, e.g. $1/(Ls + R)$ and $1/(Js + B)$ (W1D2).

The motor's two time constants: worked example



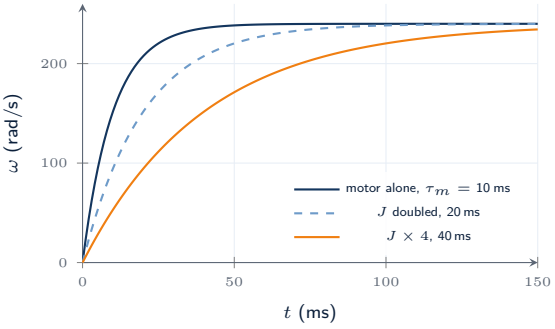
Simulated 24 V step, $R = 1\,\Omega$, $L = 0.5$ mH, $k_t = k_e = 0.1$, $J = 1 \times 10^{-4}$ kg m^2.

1. Electrical: $\tau_e = L/R = 0.5$ ms.
2. Mechanical: $\tau_m = \dfrac{RJ}{k_t k_e} = \dfrac{1 \cdot 10^{-4}}{0.01} = 10$ ms.
3. Final speed: $\omega_\infty \approx v/k_e = 240$ rad/s.
4. Start current: $\omega = 0$, no back-EMF $\Rightarrow$ up to $v/R = 24$ A (peak 21 A).

$$\tau_e = \frac{L}{R} \quad \ll \quad \tau_m = \frac{RJ}{k_t k_e}$$

Take-away: current is fast, speed is slow. Real drives limit the start current.

What the robot arm adds: load inertia slows the motor

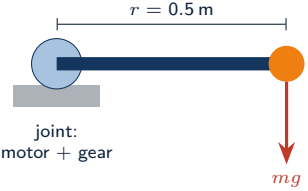


Same 24 V step, τ_e neglected: $\omega(t) = \omega_\infty(1 - e^{-t/\tau_m})$.

- The motor must accelerate its own rotor **plus** everything attached: gearbox, link, payload.
- $\tau_m \propto J_{\text{total}}$: double the inertia, double the time to speed up.
- The final speed stays the same (set by v/k_e).
- A robot arm stretched out has more inertia than when folded $\Rightarrow$ **its behaviour changes with pose**.

Gearboxes shrink the load inertia the motor "feels" by N^2 – the topic of **reflected inertia** (W1D2).

Worked example: holding a robot arm against gravity



Payload $m = 2$ kg at the end of a horizontal link (link mass neglected).

Step by step

1. Joint torque: $T_L = mgr = 2 \cdot 9.81 \cdot 0.5 \approx 9.8$ N m.
2. Direct drive, $k_t = 0.1$ N m/A: $i = 9.8/0.1 =$ **98 A** – far too much!
3. With a 100:1 gearbox the motor sees $9.8/100 = 0.098$ N m.
4. Current: $i = 0.098/0.1 \approx 1$ A; copper loss $Ri^2 \approx 1$ W ($R = 1\,\Omega$).

- Holding still needs **torque, not power**: $\omega = 0$ so $T\omega = 0$, yet current flows and heats the motor.
- That is why robot joints have gearboxes and a **holding brake**.

Session 1.3 – Check your understanding

Check your understanding

1. A motor has $\tau_m = 20\,\text{ms}$. Roughly how long until it reaches 95 % of its final speed after a voltage step?
2. You attach a load that doubles the total inertia J. What happens to τ_m and to the final speed?
3. Why is the current so large in the first milliseconds after switching on, and why does it then fall?

Hint: use the 63 % / 95 % rule, the formula $\tau_m = RJ/(k_t k_e)$ and the back-EMF $e = k_e \omega$.

Session 1.3 – Summary and key terms

- Motor = RL circuit + back-EMF, coupled to an inertia with friction and load.
- Energy conservation gives $k_t = k_e$ (SI units).
- Constant speed: straight speed–torque line; power peaks at half stall.
- Two time constants: fast electrical τ_e, slow mechanical τ_m.

Remember

$$v = Ri + L\frac{di}{dt} + k_e\omega, \quad J\frac{d\omega}{dt} = k_t i - B\omega - T_L$$
$$\tau_e = L/R, \quad \tau_m = RJ/(k_t k_e)$$

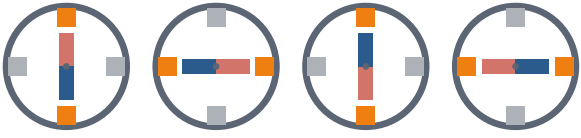
Key terms

Moment of inertia J: resistance to changes of rotation speed.
Time constant τ: time to reach 63 %.
Stiction: sticking friction at standstill.
Stall torque: torque at zero speed.
Block diagram: equations drawn as signal flow.
Disturbance: unwanted input such as T_L.

Session 4

1.4 Stepper and induction motors in automation

Stepper motor: moving in exact steps



Two phases A, B; energised poles orange. Each pulse turns the magnet rotor one step (90° here).



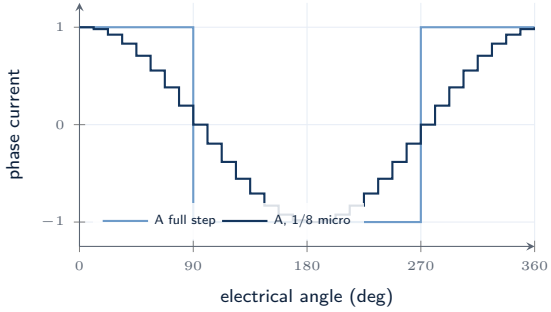
tony_duell, CC BY 2.0, via Wikimedia Commons

Analogy: a clock's second hand – it jumps tick by tick, and you know the time by counting ticks.

- Each current pulse moves the rotor **one step**.
- Real **hybrid steppers** have many teeth: **200 steps/rev** (1.8°).
- Position = number of pulses: **no sensor needed** (**open loop**).

$$\text{step angle} = \frac{360^\circ}{\text{steps per rev}}$$

Full step, half step and microstepping



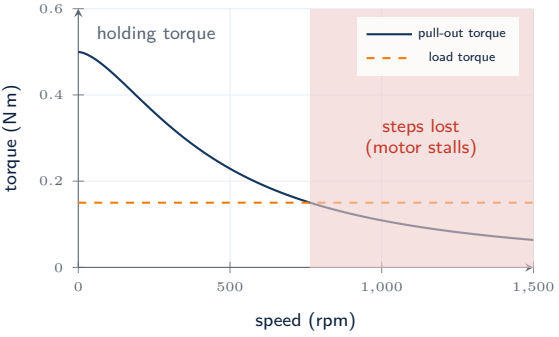
Microstepping: phase currents follow a staircase sine (phase B is the same, shifted $90°$).

- **Full step**: currents only $+I$ or $-I$; jerky, noisy.
- **Half step**: one phase at a time in between $\Rightarrow$ 400 steps/rev.
- **Microstepping**: currents shaped like sine/cosine in small stairs $\Rightarrow$ smooth, quiet.

Worked example: 3D-printer axis

1. 200 steps/rev $\times$ 16 micro $= 3200$ microsteps/rev.
2. Lead screw 8 mm/rev: $8/3200 = 2.5\,\mu$m per microstep.
3. 50 mm/s needs $50/8 \cdot 3200 = 20\,000$ pulses/s.

Stepper limits: torque falls with speed, steps can be lost

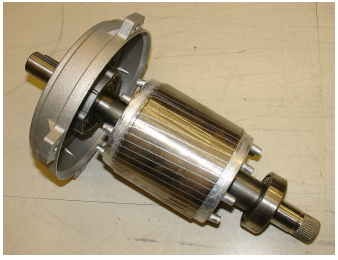


Typical shape for a NEMA-17 class motor (illustrative values).

- Winding inductance slows current rise: at high step rates the current, and so the torque, **collapses**.
- If load torque $>$ available torque, the rotor slips a step – and the controller **never knows**.
- Keep a **safety margin** (often 30–50 %) and ramp the speed up slowly.
- Holding current flows even at rest $\Rightarrow$ steppers run warm.

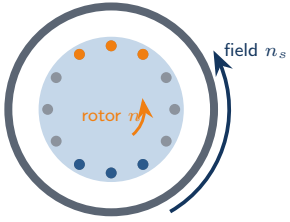
Fix: add an encoder (**closed-loop stepper**) – or use a servo motor.

Induction motor: currents induced in a cage



Zureks, CC BY-SA 3.0, via Wikimedia Commons

Squirrel-cage rotor: aluminium bars shorted by end rings, no magnets.



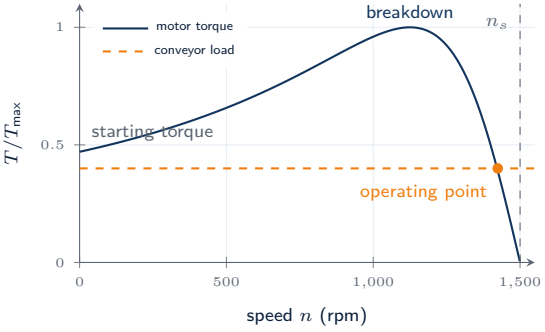
field runs ahead; bars feel it move

- Three-phase stator makes a **rotating field**.
- The field sweeps past the bars and **induces** currents in them (Faraday) $\Rightarrow$ force (Lorentz).
- Rotor runs slightly slower: **slip** s.

$$n_s = \frac{60f}{p}, \qquad s = \frac{n_s - n}{n_s}$$

Example: $f = 50\,\text{Hz}$, $p = 2$: $n_s = 1500\,\text{rpm}$; at $1440\,\text{rpm}$, $s = 4\,\%$.

Torque–speed curve of the induction motor

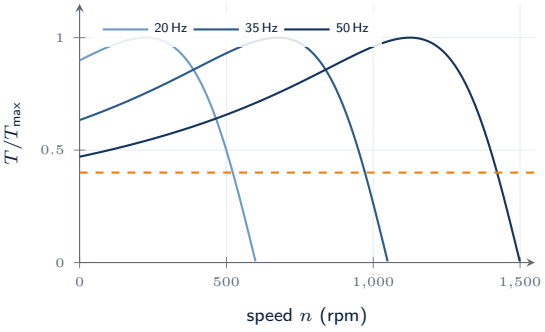


Simplified (Kloss) curve, breakdown slip 25 %.

- No slip ⇒ no induced current ⇒ **no torque** at n_s.
- Near n_s the curve is steep: speed barely changes with load (a few %).
- Load too big (past **breakdown**) ⇒ motor stalls.
- Runs straight from the mains: **cheap and rugged**.

Going further: the Kloss formula $T/T_{\max} = 2/(s/s_k + s_k/s)$ is an approximation; vector control makes an induction motor behave like a servo motor [Leonhard, 2001].

Variable speed with a frequency inverter (V/f)



Voltage scaled with frequency: same curve shape, shifted along the speed axis.

- A **frequency inverter** (VFD) makes 3-phase power at any frequency f.
- $n_s = 60f/p$ moves with $f \Rightarrow$ speed control.
- Keep V/f constant so the field (and max torque) stays the same.

Example: conveyor

4-pole motor, conveyor needs 1000 rpm: set $f \approx \frac{1000 \cdot 2}{60} +$ slip ≈ 35 Hz.

Which motor where? Choosing for an automation task

	Stepper	Induction + VFD	PMSM servo
Position feedback	none (counts steps)	usually none	**encoder**
Accuracy / dynamics	good at low speed	speed only	**excellent**
Torque at high speed	low	good	good
Cost	**low**	**low**	high
Typical task	3D printers, small gantries, lab automation	conveyors, pumps, fans, spindles	robot joints, pick-and-place, packaging

Decision questions

Must it stop at exact positions? Must it react fast? Is the load known and constant? What may it cost?

- A production line often uses **all three**.
- The robot uses servos; the conveyor feeding it an induction motor.

Session 1.4 – Check your understanding

Check your understanding

1. A stepper with 200 steps/rev drives a belt with 40 mm travel per revolution. How far does one full step move the belt?
2. The controller sent 1000 pulses, but the load jammed briefly. Where does the controller think the axis is – and where is it?
3. A 4-pole induction motor on 50 Hz turns at 1455 rpm. What is the slip?

Hint: step angle $= 360^\circ/200$; open loop means no measurement; $n_s = 60f/p$.

Session 1.4 – Summary and key terms

- Stepper: moves by pulses, no sensor; torque drops with speed; steps can be lost.
- Microstepping smooths motion and improves resolution.
- Induction: rotating field induces rotor currents; needs slip; cheap and rugged.
- A VFD changes f and so the speed; constant V/f keeps the torque.

Remember

$1.8^\circ = 200$ steps/rev, $n_s = 60f/p$, $s = (n_s - n)/n_s$

Key terms

Open loop: no measurement of the result.
Microstepping: sine-shaped staircase currents.
Pull-out torque: max torque before losing steps.
Slip: how much slower the rotor turns than the field.
Breakdown torque: max torque of an induction motor.
VFD: variable-frequency drive (inverter).

Day summary: W1D1

Energy conversion and actuators

Current + field = force; motion + field = voltage; $k_t = k_e$.
Electric for joints, pneumatic for grippers, hydraulic for heavy duty.

Motor types

DC (brushes), BLDC (6 steps, Hall), PMSM (sine, encoder), stepper (pulses), induction (slip).

Motor model

$v = Ri + L\dot{i} + k_e\omega, \quad J\dot{\omega} = k_t i - B\omega - T_L$
Fast $\tau_e = L/R$, slow $\tau_m = RJ/(k_t k_e)$.

Key numbers

$P = T\omega, \quad f_{\text{el}} = pn/60, \quad n_s = 60f/p, \quad F = pA.$

One sentence

A robot joint is an **energy converter** (motor) whose behaviour two simple ODEs describe – and whose type we choose by accuracy, dynamics and cost.

Preview of W1D2: Actuators II + Introduction to Control

Code	Session	Time
1.5	Robot joint actuators: gearboxes, reflected inertia, backlash, series-elastic	1 h
1.6	Linear actuators and end-effectors	1 h
1.7	Motor sizing and selection for a robot joint	1 h
2.1	Modelling dynamic systems	1 h
2.2	Laplace transform and transfer functions	1 h

Bring along from today:
$P = T\omega$, $T = k_t i$, the two motor equations, τ_e and τ_m, and the gravity example. Tomorrow we put a gearbox between motor and link and ask: **which motor is big enough?**

Figure credits

Slide	File (Wikimedia Commons)	Author	License
Robot joint chain	KUKA industrial robotic arm ... The Beach ... 02.jpg	Oleg Yunakov	CC BY-SA 4.0
Fluid power	A pneumatic cylinder (Max. press 1.0MPa).jpg	Jonashtand	CC BY-SA 4.0
Actuator comparison	Hydraulic cylinder.JPG	S.J. de Waard	CC BY 2.5
Brushed DC motor	Motor disassembeled.JPG	Jfer91	CC BY-SA 3.0
BLDC motor	Stator Winding of a BLDC Motor.jpg	Medvedev	CC BY-SA 3.0
PMSM	Floppy drive spindle motor open.jpg	Sebastian Koppehel	CC BY 3.0
Stepper motor	Positioner Stepper Motor parts.jpg	tony_duell	CC BY 2.0
Induction motor	Wirnik by Zureks.jpg	Zureks	CC BY-SA 3.0

Source pages: `https://commons.wikimedia.org/wiki/File:<file name>` (see `figures/CREDITS.tsv`).

All other drawings, plots and diagrams: own work for this course (TikZ / pgfplots / circuitikz).

References

- A. E. Fitzgerald, C. Kingsley, S. D. Umans, *Electric Machinery*, 6th ed., McGraw-Hill, 2003.
- A. Hughes, B. Drury, *Electric Motors and Drives*, 4th ed., Newnes, 2013.
- R. Krishnan, *Electric Motor Drives: Modeling, Analysis, and Control*, Prentice Hall, 2001.
- R. Krishnan, *Permanent Magnet Synchronous and Brushless DC Motor Drives*, CRC Press, 2010.
- N. Mohan, *Electric Drives: An Integrative Approach*, MNPERE, 2003.
- W. Leonhard, *Control of Electrical Drives*, 3rd ed., Springer, 2001.
- T. Kenjo, A. Sugawara, *Stepping Motors and Their Microprocessor Controls*, 2nd ed., Oxford Univ. Press, 1994.
- B. Siciliano, L. Sciavicco, L. Villani, G. Oriolo, *Robotics: Modelling, Planning and Control*, Springer, 2009.
- J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 4th ed., Pearson, 2018.
- W. Bolton, *Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering*, 7th ed., Pearson, 2019.
- B. Armstrong-Hélouvry, P. Dupont, C. Canudas de Wit, “A survey of models, analysis tools and compensation methods for the control of machines with friction”, *Automatica* 30(7), 1994.
- IEC 60034-1, *Rotating electrical machines – Part 1: Rating and performance*; IEC 61800 series, *Adjustable speed electrical power drive systems*.